

Technical Datasheet Input / Output Modules with Modbus RTU Protocol with RS485 Interface

The IO modules communicate via RS485. The port can drive distances up to max 700 meters without the use of any repeater (*this feature however also depends on the signal strength of the Modbus Master Device*).

The RS485 Digital IO module is sturdy, low power usage and easy to use.

8 Port AI Module: -



The IO modules are mounted on DIN rail mountable casing and with exposed connectors and LED indicators. The DIP switch for Slave ID and Baud rate are placed inside the enclosure.

The design of the modules incorporates '**resettable Fuses**' to safeguard against sudden High Voltage spikes. The board also has '**reverse polarity**' protection both for input **Power** connector.

The communication port also has signal isolation and additional Jumper for 'end of line' 120 ohms termination resistance which may or may not be used.

Specifications

General

I/O Connectors	2 Pin 5.08 mm pitch pluggable screw terminals.
Dimensions	70 mm L x 110 mm B x 50 mm H
Power	Input Power – 12 – 24 VDC Typical – 12V DC @ 1 Amp
Operating Temperature	0 – 60° C (32 ~ 140°F)
Storage Temperature	-20 - 70° C (-4 ~ 158°F))
Storage Humidity	5 ~ 95 % RH, non – Condensing

Certifications



AI Inputs –

Channels	8
Input Signal	4 – 20 mA / 0 – 5 V / 0 – 10 V / 0 – 5 V (jumper sel)
Accuracy	± 2 % of Full scale
Input Resolution	10-bit /12-bit resolution (optional)
Isolation	Optically Isolated
External Loop Voltage	+ 12 VDC min
AI input impedance	120Ω

Additional Features: -

Communication port isolated
Input power reverse polarity safety
ESD Safety IEC 61000-4-2, ± 30KV contact, ± 30KV
airEFT IEC 61000-4-4, 50A (5/50ms)
750V isolation.
CRC Error check.
No configuration needed on the IO board

Configuration Settings: -

Communication Speed	9600 – 19200 Kbps (DIP SW Selectable)
Data Bits	8
Parity	None
Stop bit	1
CRC	Yes
Slave ID	1 – 15 (DIP SW Selectable)
Function Code AI	0x09 Read Holding Registers
AI Register Address	10 Bit - 0,1,2,3,4,5,6,7 / 12 Bit – 8,9,10,11,12,13,14,15

ID	Function Description	Register Description	Modbus Function Code	Protocol	Data Type
1	AI 1 – 10 Bit	40001	0X03	RS485	16 Bit Unsigned int
2	AI 2 – 10 Bit	40002	0X03	RS485	16 Bit Unsigned int
3	AI 3 – 10 Bit	40003	0X03	RS485	16 Bit Unsigned int
4	AI 4 – 10 Bit	40004	0X03	RS485	16 Bit Unsigned int
5	AI 5 – 10 Bit	40005	0X03	RS485	16 Bit Unsigned int
6	AI 6 – 10 Bit	40006	0X03	RS485	16 Bit Unsigned int
7	AI 7 – 10 Bit	40007	0X03	RS485	16 Bit Unsigned int
8	AI 8 – 10 Bit	40008	0X03	RS485	16 Bit Unsigned int
9	AI – 1 12 Bit	40009	0x03	RS485	16 Bit Unsigned int
10	AI – 2 12 Bit	40010	0x03	RS485	16 Bit Unsigned int
11	AI – 3 12 Bit	40011	0x03	RS485	16 Bit Unsigned int
12	AI – 4 12 Bit	40012	0x03	RS485	16 Bit Unsigned int
13	AI – 5 12 Bit	40013	0x03	RS485	16 Bit Unsigned int
14	AI – 6 12 Bit	40014	0x03	RS485	16 Bit Unsigned int
15	AI – 7 12 Bit	40015	0x03	RS485	16 Bit Unsigned int
16	AI – 8 12 Bit	40016	0x03	RS485	16 Bit Unsigned int

Note: -

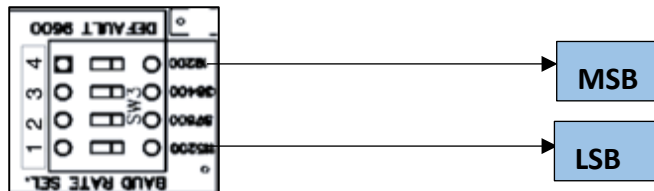
For MODBUS communications, a **shielded and twisted pair cable** is used. One example of such cable is Belden 3105A.

Recommended Cable Electrical Characteristics: -

22 AWG Cable	Shielded and twisted pair should be used.
Tinned Copper	Recommended
Nominal Conductor DCR	14.7 ohm / 1000 ft
Nominal Capacitance	11 pf / feet (conductor to conductor)
High Frequency Non-Insertion Loss	0.5db / 100ft

NOTE – By default the Jumper setting on the board is 10 V and 10 Bit.

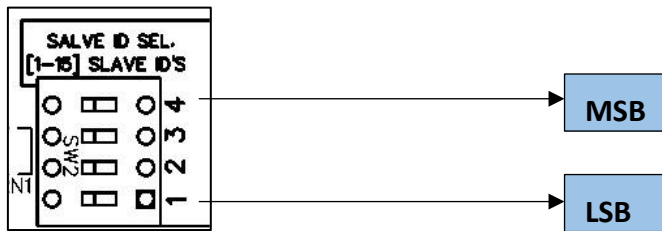
BAUD RATE DESCRIPTION



- For Baud rate Selection, DIP SW is used as per the diagram.
- Pulling up the switch will make Baud rate active.
- If no selection is made 9600 will be default Baud rate.
- When u change the Baud rate in the Module power 'ON' condition, please press the reset button to get Change to affect.

Baud Rate	DIP SWITCH			
	1	2	3	4
9600	OFF	OFF	OFF	OFF
115200	ON	OFF	OFF	OFF
57600	OFF	ON	OFF	OFF
38400	OFF	OFF	ON	OFF
19200	OFF	OFF	OFF	ON

SLAVE ID DESCRIPTION



For Slave ID Selection SW is used to Set The SLAVE ID .

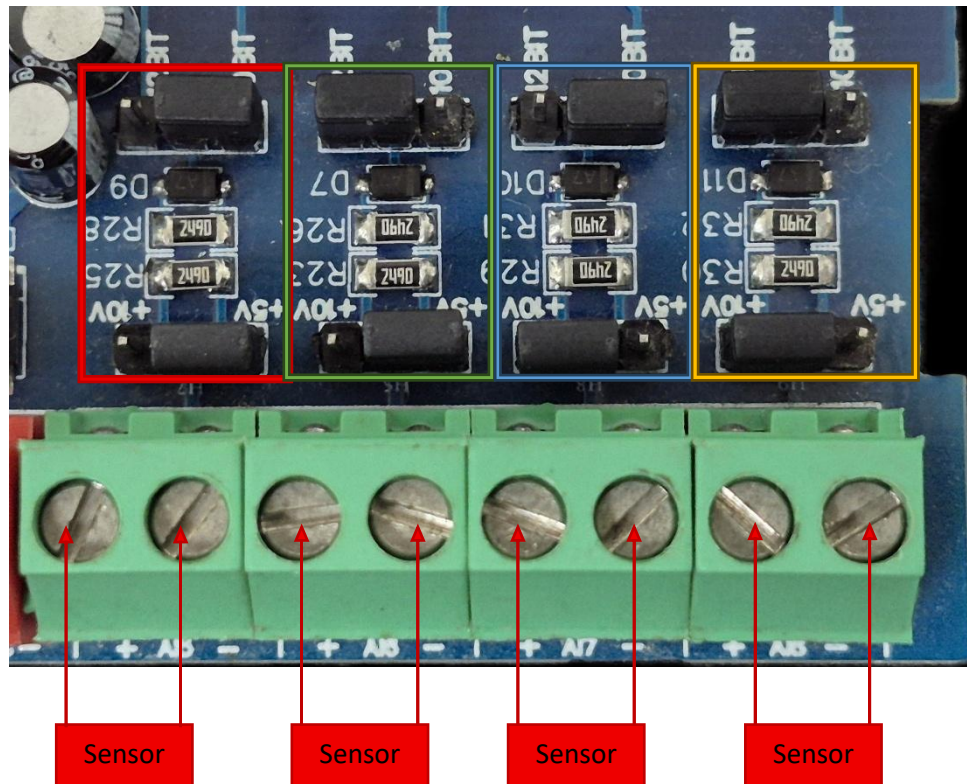
For Slave ID DIP Switch **LSB is "1"** follow through **"4" is MSB.**

Slave ID Confirmed through below Device ID table .

IF Eg. Slave ID 1 is Needed to be selected Switch number 1 should pulled up other three should be selected down side. So"1 0 0 0" will be selected as Slave ID 1.

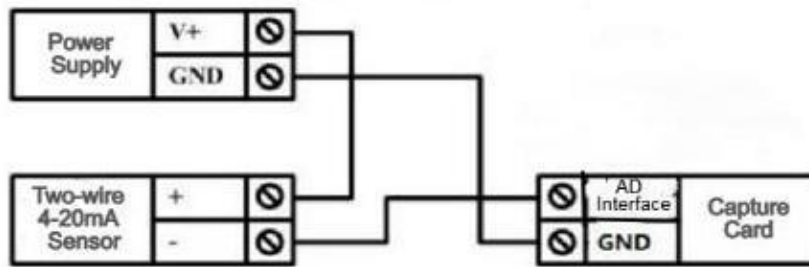
Slave ID	DIP SWITCH				OUTPUT (Binary)	OUTPUT (Decimal)
	1	2	3	4		
0	OFF(0)	OFF(0)	OFF(0)	OFF(0)	0 0 0 1	1
1	ON(1)	OFF(0)	OFF(0)	OFF(0)	0 0 0 1	1
2	OFF(0)	ON(1)	OFF(0)	OFF(0)	0 0 1 0	2
3	ON(1)	ON(1)	OFF(0)	OFF(0)	0 0 1 1	3
4	OFF(0)	OFF(0)	ON(1)	OFF(0)	0 1 0 0	4
5	ON(1)	OFF(0)	ON(1)	OFF(0)	0 1 0 1	5
6	OFF(0)	ON(1)	ON(1)	OFF(0)	0 1 1 0	6
7	ON(1)	ON(1)	ON(1)	OFF(0)	0 1 1 1	7
8	OFF(0)	OFF(0)	OFF(0)	ON(1)	1 0 0 0	8
9	ON(1)	OFF(0)	OFF(0)	ON(1)	1 0 0 1	9
10	OFF(0)	ON(1)	OFF(0)	ON(1)	1 0 1 0	10
11	ON(1)	ON(1)	OFF(0)	ON(1)	1 0 1 1	11
12	OFF(0)	OFF(0)	ON(1)	ON(1)	1 1 0 0	12
13	ON(1)	OFF(0)	ON(1)	ON(1)	1 1 0 1	13
14	OFF(0)	ON(1)	ON(1)	ON(1)	1 1 1 0	14
15	ON(1)	ON(1)	ON(1)	ON(1)	1 1 1 1	15

AI

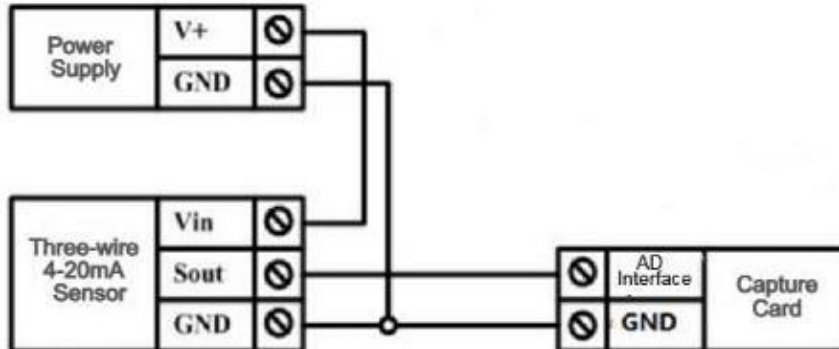


- **Analog Input (AI) Operation:** Each channel provides two terminals: + (Positive) and - (Negative/Common). These are used to read external analog sensor signals.
- **Channel Configuration (Visual Guide):** Each of the four AI channels features two dedicated jumpers to independently set the voltage range and bit resolution. Refer to the color-coded boxes in the board diagram to configure your desired scenario:
 - **Red Box (5V, 10-bit):** Set the voltage jumper to **+5V** and the resolution jumper to **10 BIT**.
 - **Green Box (5V, 12-bit):** Set the voltage jumper to **+5V** and the resolution jumper to **12 BIT**.
 - **Blue Box (10V, 10-bit):** Set the voltage jumper to **+10V** and the resolution jumper to **10 BIT**.
 - **Yellow Box (10V, 12-bit):** Set the voltage jumper to **+10V** and the resolution jumper to **12 BIT**.
- **Sensor Wiring Connections:**
 - The analog sensor's positive signal wire should be connected to the **+** terminal of the desired channel.
 - The sensor's common or ground wire should be connected to the corresponding **-** terminal.
- **Crucial Setup Recommendation:** Always ensure that the voltage jumper setting strictly matches the maximum output range of your connected sensor (either 5V or 10V) to guarantee accurate analog-to-digital conversion and prevent data clipping.

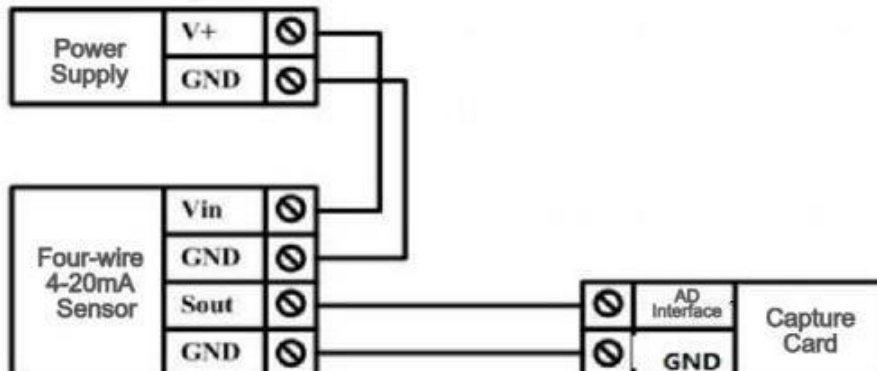
Two-wire Sensor Wiring Diagram

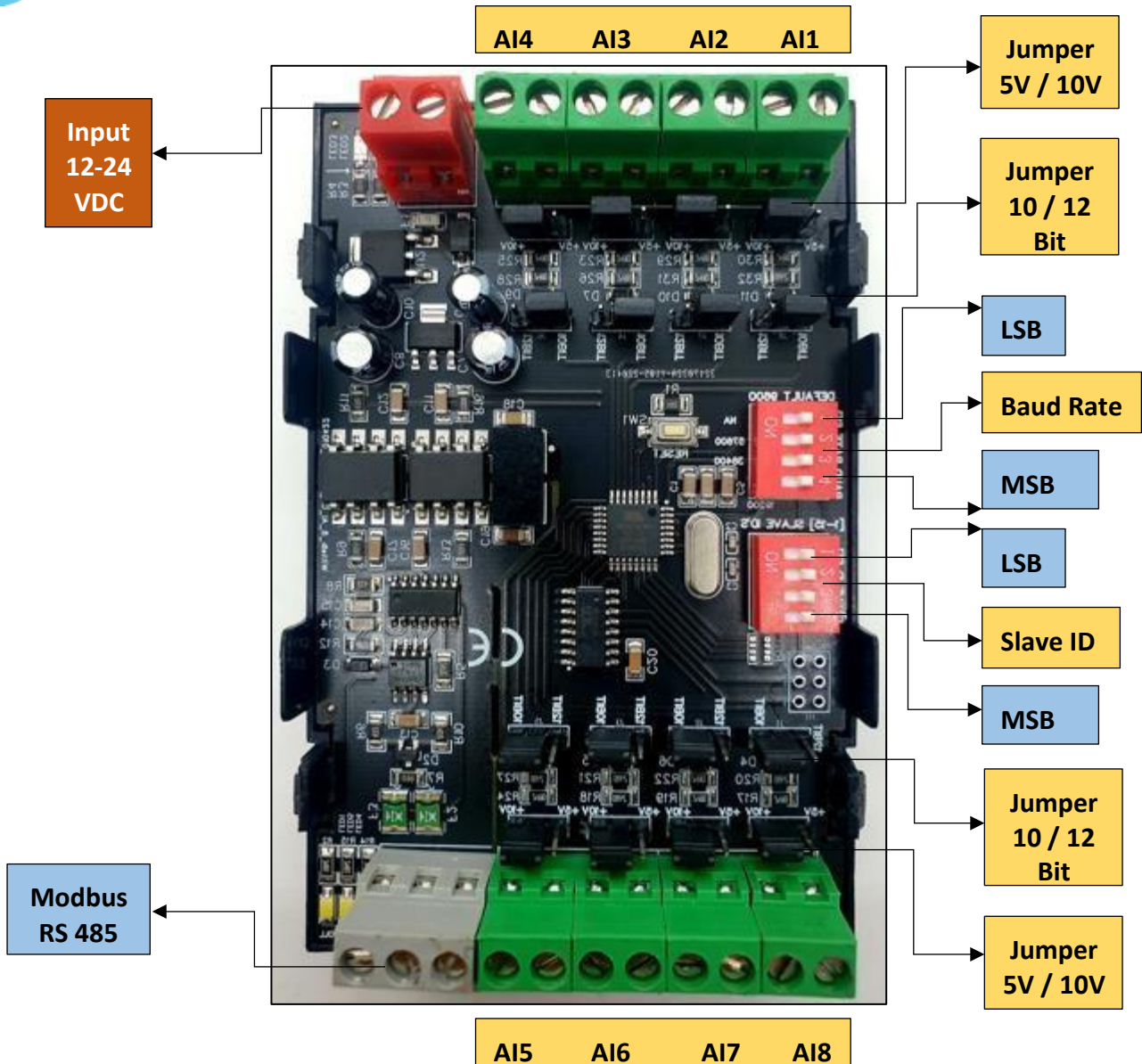


Three-wire Sensor Wiring Diagram



Four-wire 4-20mA Sensor





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